

# Yahya SALEH

## CURRICULUM VITAE

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**Research interests.** Approximation theory, scientific machine learning, statistical learning theory, computational quantum molecular physics.

## EMPLOYMENT

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| since 04/2025     | Consultant, <b>d-fine GmbH</b>                                                                                                     |
| 04/2023 – 03/2025 | Postdoctoral Researcher, Department of Mathematics, <b>Universität Hamburg</b>                                                     |
| 04/2020 – 03/2023 | Doctoral Researcher, Department of Mathematics, <b>Universität Hamburg and Center for Free-Electron Laser Science (CFEL), DESY</b> |
| 05/2018 – 09/2018 | Research Assistant, <b>Center for Free-Electron Laser Science (CFEL), DESY</b>                                                     |

## EDUCATION

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| 04/2020 – 05/2023 | <b>Dr. rer. nat. in Mathematics</b> ( <i>magna cum laude</i> ), Universität Hamburg<br>Graduate school: Data Science in Hamburg – Helmholtz Graduate School for the Structure of Matter (DASHH)<br>Thesis: <i>Spectral and active learning for enhanced and computationally scalable quantum molecular dynamics</i><br>Supervisors: Prof. Dr. Armin Iske, Prof. Dr. Jochen Küpper, Dr. Andrey Yachmenev |
| 2017 – 10/2019    | <b>M.Sc. in Mathematical Modelling in Engineering (MathMods)</b> , joint <b>Erasmus Mundus degree</b> , Università degli Studi dell'Aquila, Universität Hamburg, Universitat Autònoma de Barcelona<br>Thesis: <i>On non-adiabatic dynamics of deflected molecular beams</i><br>Supervisors: Prof. Dr. Armin Iske, Prof. Dr. Jochen Küpper, Dr. Andrey Yachmenev                                         |
| 2012 – 09/2017    | <b>B.Eng. in Electrical Engineering (Electronics and Telecommunications)</b> , Damascus University<br>Thesis: <i>Designing a cellular-based tracking system using machine learning and Kalman filtering</i><br>Supervisors: Prof. Dr. Hiyam Khaddam                                                                                                                                                     |

## TEACHING EXPERIENCE

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Universität Hamburg, Department of Mathematics

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| Winter 2024/25 | <i>Software Practicum</i> (M.Sc.) — Lecturer, lectures and exercise sessions, in German                                                              |
| Summer 2024    | <i>Mathematical Machine Learning</i> (M.Sc.) — Lecturer, lectures and exercise sessions, in English                                                  |
| Winter 2023/24 | <i>Higher Analysis</i> (B.Sc.) — Teaching assistant to Prof. Dr. Armin Iske, exercise sessions, in German                                            |
| Summer 2023    | <i>Numerical Methods for Partial Differential Equations</i> (M.Sc.) — Teaching assistant to Prof. Dr. Hendrik Ranocha, exercise sessions, in English |

## STUDENTS MENTORED

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| 2024–2025 | Sharlet Shaji, Master's student, Universität Hamburg, Department of Mathematics — Spectral learning for solving the time-dependent Schrödinger equation; co-supervised with Prof. Dr. Armin Iske |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

2021–2022 Jannik Eggers, Master's student, Universität Hamburg, Department of Mathematics —  
Solution of the vibrational Schrödinger equation using neural networks; co-supervised  
with Dr. Jens-Peter M. Zemke and Prof. Dr. Jochen Küpper

## RESEARCH STAYS AND GRANTS

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08/2024 – Research stay with Prof. Dr. Martin Frank, Karlsruhe Institute of Technology (KIT),  
09/2024 funded by a research-stay grant of the Helmholtz Information and Data Science  
Academy (HIDA)

## PUBLICATIONS

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Reverse chronological order. Key publications are marked with \*.

### Peer-reviewed articles

- [1] E. Vogt, Á. F. Corral, **Y. Saleh**, “Adaptive Vibrational Coordinates via Symmetry-Aware Normalizing Flows,” *Journal of Chemical Theory and Computation* **22**(12), 6137 (2026). doi:10.1021/acs.jctc.6c00224
- \* [2] **Y. Saleh**, A. Iske, “Inducing Riesz Bases in  $L^2$  via Composition Operators,” *Complex Analysis and Operator Theory* **20**, 21 (2026). doi:10.1007/s11785-025-01874-5
- [3] A. Yachmenev, E. Vogt, Á. F. Corral, **Y. Saleh**, “Taylor-mode automatic differentiation for constructing molecular rovibrational Hamiltonian operators,” *The Journal of Chemical Physics* **163**(7), 072501 (2025). doi:10.1063/5.0287347
- [4] E. Vogt, Á. F. Corral, **Y. Saleh**, A. Yachmenev, “Transferability and interpretability of vibrational normalizing-flow coordinates,” *The Journal of Chemical Physics* **163**(15), 154106 (2025). doi:10.1063/5.0285954
- \* [5] **Y. Saleh**, Á. F. Corral, E. Vogt, A. Iske, J. Küpper, A. Yachmenev, “Computing Excited States of Molecules Using Normalizing Flows,” *Journal of Chemical Theory and Computation* **21**(10), 5221–5229 (2025). doi:10.1021/acs.jctc.5c00590
- \* [6] V. Menden, **Y. Saleh**, A. Iske, “Bounds on the Generalization Error in Active Learning,” *Proceedings of the 6th Northern Lights Deep Learning Conference (NLDL)* **265**, 168–175 (2025). proceedings.mlr.press/v265/menden25a.html
- [7] T. Wenzel, **Y. Saleh**, A. Iske, “Two-Layered and Deep Kernels as Data-Adapted Kernels,” *DEEPK 2024: International Workshop on Deep Learning and Kernel Machines* (2024). www.esat.kuleuven.be/stadius/E/DEEPK2024/7\_two\_layered\_and\_deep\_kernels\_a.pdf
- [8] **Y. Saleh**, A. Iske, A. Yachmenev, J. Küpper, “Augmenting Basis Sets by Normalizing Flows,” *Proceedings in Applied Mathematics and Mechanics* **23**(1), e202200239 (2023). doi:10.1002/pamm.202200239
- \* [9] **Y. Saleh**, V. Sanjay, A. Iske, A. Yachmenev, J. Küpper, “Active learning of potential-energy surfaces of weakly bound complexes with regression-tree ensembles,” *The Journal of Chemical Physics* **155**(14), 144109 (2021). doi:10.1063/5.0057051

### Preprints

- \* [10] **Y. Saleh**, “Convergence theory for Hermite approximations under adaptive coordinate transformations,” arXiv:2604.16975 [math.NA] (2026).
- [11] Á. F. Corral, **Y. Saleh**, “Enhancing polynomial approximation of continuous functions by composition with homeomorphisms,” arXiv:2512.13740 [math.NA] (2025).

## CONTRIBUTIONS TO INTERNATIONAL CONFERENCES AND WORKSHOPS

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### Invited talks

2026 “Learning, sharing, and using optimized vibrational coordinates,” ExoMol 2026,  
University College London, United Kingdom

2025 “Normalizing flows: from learning probability distributions to basis-set discovery,”  
Linköping University, Sweden

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|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024                                   | "Learning basis sets in $L^2$ and application to computing excited states of molecules," <a href="#">BIRS Workshop on Modern Methods for Differential Equations of Quantum Mechanics</a> , Banff, Canada           |
| 2023                                   | "Flow-induced bases and application to quantum molecular physics," Machine Learning in Engineering, TU Hamburg, Germany                                                                                            |
| 2023                                   | "Augmenting bases with normalizing flows for solving Schrödinger equations," DPG SAMOP Spring Meeting, Germany                                                                                                     |
| 2021                                   | "Applications of machine learning in quantum simulations of hydrogen bond dynamics," <a href="#">Machine Learning for Quantum X</a> , online                                                                       |
| <i>Contributed talks</i>               |                                                                                                                                                                                                                    |
| 2026                                   | "Learning bases via normalizing flows, applications to solving molecular Schrödinger equations and image compression," International Conference on Scientific Computing and Machine Learning, Bath, United Kingdom |
| 2025                                   | "Learning basis sets using unitary and bounded bijective operators," International Conference in Numerical Mathematics and Scientific Computing, Uppsala University, Sweden                                        |
| 2024                                   | "Spectral learning for solving molecular Schrödinger equations," <a href="#">PINN-PAD: Physics-Informed Neural Networks in Padova</a> , Padova, Italy                                                              |
| 2024                                   | "Augmenting spectral methods with invertible neural networks and application to quantum molecular physics," <a href="#">ALGORITMY 2024</a> , Podbanské, Slovakia                                                   |
| 2022                                   | "Spectral learning for solving the Schrödinger equation for molecules," Computational Methods in Applied Mathematics (CMAM), TU Wien, Austria                                                                      |
| 2022                                   | "Deep spectral methods for solving variational problems arising from differential equations," GAMM Annual Meeting, RWTH Aachen, Germany                                                                            |
| 2022                                   | "Deep spectral methods for solving variational problems arising from differential equations," HCM Workshop: Synergies Between Data Science and PDE Analysis, Universität Bonn, Germany                             |
| 2022                                   | "Spectral learning for (ro-)vibrational calculations of weakly bound molecules," DPG SAMOP Spring Meeting, Germany                                                                                                 |
| 2021                                   | "Active learning for computing potential-energy surfaces of molecules," Bunsentagung, online                                                                                                                       |
| <i>Posters</i>                         |                                                                                                                                                                                                                    |
| 2025                                   | "Bounds on the generalization error in active learning," 6th Northern Lights Deep Learning Conference (NLDL), Tromsø, Norway                                                                                       |
| 2024                                   | "Computing excited states of molecules using normalizing flows," High-Dimensional Quantum Dynamics Conference, Hamburg, Germany                                                                                    |
| 2022                                   | "Deep learning for computing spectra of molecules," Opening Symposium, Center for Data and Computing in Natural Sciences (CDCS), Hamburg, Germany                                                                  |
| 2022                                   | "Deep learning for computing rovibrational spectra of molecules," European CFEL and DESY Photon Science Users' Meeting, Hamburg, Germany                                                                           |
| 2021                                   | "Applications of machine learning in quantum simulations of hydrogen bond dynamics," Warsaw Summer School for Quantum Physics and Chemistry, University of Warsaw, Poland                                          |
| <i>Schools and hackathons attended</i> |                                                                                                                                                                                                                    |
| 2022                                   | Hausdorff School: Foundational Methods in Machine Learning, Universität Bonn, Germany                                                                                                                              |
| 2021                                   | Helmholtz H3 Hackathon                                                                                                                                                                                             |

## LANGUAGES AND SKILLS

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|-------------|---------------------------------------------------------------------------|
| Languages   | Arabic (native) · English (C2) · German (C2) · Italian (C1) · French (B2) |
| Programming | Python (JAX, NumPy, SciPy), Julia, C++, MATLAB, Java, Bash                |
| Tools       | Git, GitHub/GitLab, CI/CD (Jenkins), Docker, Kubernetes                   |